

Assignment – 3.4

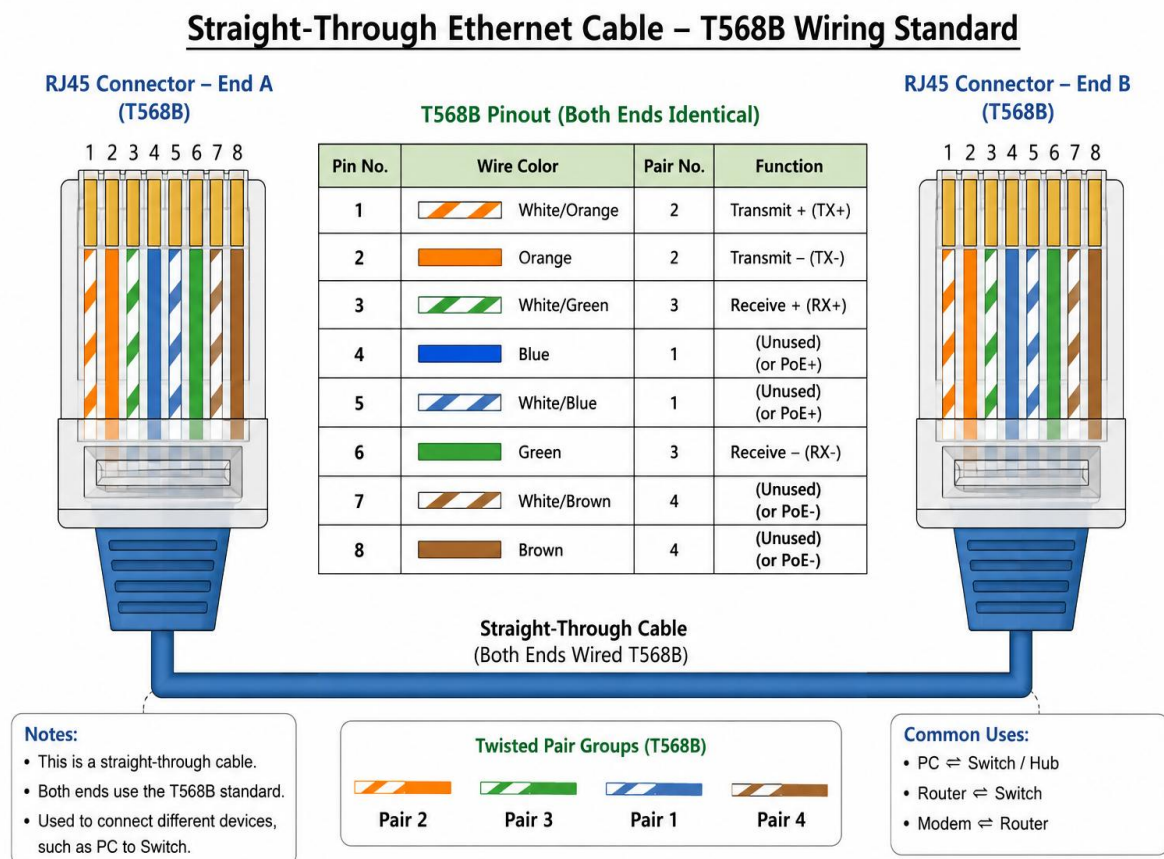
Q - 1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Difference	Ethernet Cable	Fiber Optic Cable
Signal	Uses electrical signals.	Uses light signals.
Material	Uses copper wires	Uses glass/plastic fibers
Speed & distance	Good speed, usually for shorter distances	Very high speed and works over much longer distances
Cost	Cheaper and easier to install	More expensive and needs special tools

Real-world examples:

- **Ethernet cable:** Connecting a home Wi-Fi router to a computer or smart TV.
- **Fiber optic cable:** Connecting buildings in a college campus backbone or data center.

Q - 2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



Q - 3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Attaching an RJ45 Connector to a CAT6 Cable

Tools needed:

- CAT6 cable
- RJ45 connector
- Crimping tool
- Cable cutter/stripper
- Cable tester (optional)

Steps:

1. **Cut the cable** – Cut the CAT6 cable to the required length.
2. **Remove the outer cover** – Strip about 2–3 cm of the outer jacket.
3. **Separate the wires** – Untwist the four pairs of small wires.
4. **Arrange the wires** – For **T568B**, arrange them in this order:
 - White-Orange
 - Orange
 - White-Green

- Blue
- White-Blue
- Green
- White-Brown
- Brown

5. **Straighten the wires** – Make all 8 wires straight and keep them in the correct order.
6. **Cut the ends evenly** – Trim the wires so they are the same length.
7. **Insert into RJ45** – Put all 8 wires inside the RJ45 connector. Make sure they reach the end.
8. **Crimp the connector** – Put the connector into the crimping tool and press it firmly.
9. **Check the cable** – Make sure the connector is tight and the wires are in the correct order.
10. **Test it** – Use a cable tester to check if the cable works correctly.

**Q - 4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.
Hint: Consider speed, distance, and budget for your answer.**

- **Cable type:** I would use **Ethernet cable (CAT6)** for the 10 gaming PCs.
- **Tool:** I would use an **RJ45 crimper** to attach RJ45 connectors to the Ethernet cables.
- **Reason:** Ethernet is fast enough for LAN gaming, works well over the short distances in a café, and is cheaper than fiber optics.
- **Budget:** CAT6 cables and an RJ45 crimper are affordable and easier to install and maintain.